

Walmart Sales Analysis Report

Project Overview

This project analyzes Walmart's transactional sales data to uncover revenue patterns, customer behavior, branch performance, and operational trends. The analysis follows an end-to-end data analytics workflow using **Python, SQL, and Tableau**, resulting in an executive-ready dashboard with actionable business insights.

Data Source

- **Source:** Kaggle (Walmart Sales Dataset)
 - **Time Period:** 2019–2023
 - **Records:** 9,969 transactions
 - **Key Attributes:** Branch, City, Category, Date & Time, Quantity, Revenue, Payment Method, Customer Rating
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Data Collection & Preparation

- Data was downloaded from Kaggle and imported into **VS Code**.
 - Cleaning and preprocessing were performed using **Python (Pandas)**:
 - Corrected date parsing issues and standardized datetime formats
 - Split date and time into analytical components (year, month, day, hour)
 - Ensured numeric consistency for revenue, quantity, and ratings
 - Created derived metrics such as **Average Basket Value**
 - Cleaned data was stored in **PostgreSQL** for structured analysis.
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SQL Analysis

SQL was used to answer multiple business questions using:

- Aggregations and grouping
- Ranking and window functions
- Year-over-year revenue comparisons
- Time-based and categorical analysis

Key analyses included:

- Branch-level revenue performance
 - Revenue decline percentage by branch
 - Peak sales hours and busiest weekdays
 - Payment method adoption
 - Category-wise revenue and profitability
 - City and category-based customer rating patterns
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Tableau Dashboard

An interactive Tableau dashboard was built to present insights clearly to decision-makers.

Executive KPIs

- **Total Transactions:** 9,969
- **Total Revenue:** \$1.21M
- **Average Rating:** 5.83 / 10
- **Average Basket Value:** \$121.35

Dashboard Design

- No global slicers to avoid KPI distortion
 - **Revenue by Branch** chart acts as the primary interactive filter
 - Clean layout focused on clarity and executive consumption
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Key Business Insights

- **Overall Performance**
 - \$1.21M revenue generated from **9,969 transactions**
 - Each transaction contributes **\$121.35 on average**, indicating high per-customer spend
- **Branch Performance**
 - Top-performing branches generate **~15–20% more revenue** than the lowest-performing ones

- Revenue is concentrated among a small set of branches, indicating uneven performance distribution
 - **Revenue Trends**
 - Year-end periods show revenue spikes exceeding **\$60K per month**
 - Sharp drops followed by recoveries indicate **seasonal demand and possible operational disruptions**
 - **Customer Timing Behavior**
 - **Afternoon and evening hours contribute ~65% of total transactions**
 - **3 PM–7 PM alone accounts for over 40% of daily sales volume**
 - **Weekly Sales Pattern**
 - **Tuesday and Thursday** are the busiest days, each contributing **~14–15% of weekly transactions**
 - **Monday is the weakest day**, with **~13% fewer transactions** than peak weekdays
 - **Payment Method Usage**
 - **Credit cards:** ~42.7% of payments
 - **E-wallets:** ~38.9%
 - **Cash:** ~18.4%, showing strong customer shift toward digital payments
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Business Recommendations

- **Improve Underperforming Branches**
 - Prioritize branches generating **15–20% lower revenue** than top performers
 - Replicate operational strategies from high-performing locations
- **Optimize Staffing**
 - Increase staffing during **3–7 PM**, which drives **40%+ of daily transactions**
- **Promote Digital Payments**
 - Focus incentives on credit cards and e-wallets, already used in **80%+ of transactions**
- **Enhance Customer Experience**

- Improve service quality to raise average ratings by **1–1.5 points**, which can significantly impact repeat sales
 - **Stabilize Revenue Volatility**
 - Address causes of sharp revenue drops to reduce dependency on seasonal peaks
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Tools & Technologies

- **Python:** Pandas
 - **Database:** PostgreSQL
 - **SQL:** Aggregations, Window Functions, CTEs
 - **Visualization:** Tableau
 - **Environment:** VS Code, Jupyter Notebook
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Conclusion

This project demonstrates a complete analytics pipeline—from raw data acquisition and cleaning to SQL-driven insights and executive-level dashboarding. The analysis highlights key revenue drivers, customer behavior patterns, and operational opportunities, enabling data-driven decision-making for business optimization.
